

SMART ADDA – FYP PROJECT DOCUMENTATION

1. Project Title

Smart Adda – Bus Seat Booking and Coach Tracking Mobile Application

2. Introduction

Smart Adda is a mobile application developed to make bus travel easier for passengers and transport companies. In many areas, people still go physically to the bus adda to book seats. This process takes time and sometimes passengers cannot get accurate information about schedules or bus locations.

The Smart Adda application allows passengers to book bus seats using their mobile phones. They can also track the coach location, view schedules, and get digital tickets.

The system also helps transport companies manage their buses, drivers, and passenger bookings through an admin panel.

The application is developed using **Flutter for the frontend and Firebase for the backend services.**

3. Problem Statement

In traditional bus booking systems, passengers face several problems:

- Long queues at bus terminals (Adda)
- No online seat booking system
- Difficulty in knowing the exact bus departure time
- No tracking system for buses
- Manual management of passengers and schedules

Transport companies also face problems managing drivers, routes, and bookings manually.

Smart Adda aims to solve these problems by providing a **digital and automated platform for bus booking and tracking.**

4. Objectives of the Project

The main objectives of Smart Adda are:

- To provide an **easy seat booking system**
- To allow passengers to **view bus schedules**
- To provide **live tracking of coaches**

- To generate **digital tickets**
- To help companies **manage drivers, buses, and routes**
- To reduce time and effort for passengers

5. Scope of the Project

The Smart Adda application will include the following features:

Passenger Features

- View nearby Adda locations
- View available coaches
- Check schedules
- Book seats online
- Make online payment
- Receive digital ticket
- Track bus location
- View driver details

Driver Features

- View assigned route
- Update coach location
- Manage trip information
- View passenger list

Admin Features

- Add and manage drivers
- Add buses and routes
- Manage schedules
- Monitor bookings
- View passenger information

6. System Modules

The system consists of **three main modules**:

1. Passenger Module

Passengers can:

- Register and login
- View available buses

- Book seats
- Make payments
- Track bus location
- View e-tickets

2. Driver Module

Drivers can:

- Login to the system
- View trip details
- Share live location
- View passenger information

3. Admin Module

Admin can:

- Manage routes
- Manage drivers
- Manage vehicles
- Monitor bookings
- Control the overall system

7. Methodology

The Smart Adda system is developed using a **mobile application with cloud backend services**.

1. Frontend (Mobile App)

The user interface is developed using **Flutter** which allows the application to run on Android devices.

2. Backend (Firebase)

Firebase is used to manage:

- User authentication
- Database storage
- Booking information
- Live location updates

3. GPS Tracking

The application uses **mobile GPS location** to track the coach location.

4. Google Maps API

Google Maps API is used to:

- Show coach location
- Display routes
- Show Adda locations

8. System Features

The application provides the following features:

1. Adda Selection

Passengers can select the nearest bus adda.

2. Vehicle Details

Users can see coach details such as:

- Bus number
- Driver information
- Seat availability

3. Seat Booking

Passengers can select and book seats through the mobile app.

4. Online Payment

Passengers can pay through:

- QR payment
- Cash option

5. Live Tracking

Passengers can track the coach location using GPS.

6. Digital Ticket

After booking, passengers receive a **digital ticket with QR code**.

7. Chat Support

Passengers can communicate with support or drivers through chat.

9. Tools and Technologies

Category	Tools / Technologies
Frontend	Flutter
Backend	Node.js / Express.js
Database	Firebase
Cloud Storage	Firebase Cloud
Maps	Google Maps API
Design	Figma
IDE	Android Studio
AI Integration	OpenAI API

10. Expected Results

After completing the Smart Adda project:

- Passengers will be able to **book bus seats easily through mobile app**
- Transport companies will manage **drivers, routes, and bookings digitally**
- Passengers can **track buses in real time**
- The system will reduce manual work and waiting time

11. Conclusion

Smart Adda is a smart transportation solution that simplifies the process of bus seat booking and tracking. The application will improve convenience for passengers and efficiency for transport companies. By using modern technologies like Flutter, Firebase, and GPS tracking, the system will provide a fast and reliable travel booking platform.